

CSE 3112 Midterm 3

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I. QUESTION (20 POINTS)

Given the adjacency matrix representation of a graph G , return the incoming nodes to a given node i as a linked list.

II. QUESTION (20 POINTS)

Modify breadth first search algorithm such that it will calculate the shortest path from the source node to all nodes in an unweighted graph.

III. QUESTION (20 POINTS)

There are n men who need to be married to n women, one man to one woman. The happiness that would occur if the i th man is married to j th woman is given in a matrix C_{ij} . Write a greedy algorithm to find an assignment with the maximum happiness. Your function will return an array of man indexes which are married to the women.

IV. QUESTION (20 POINTS)

Modify depth first search algorithm such that it will determine if the undirected graph can be colored with just two colors so that no two adjacent vertices are in the same color.

V. QUESTION (20 POINTS)

The swap sorting of permutation π is a transformation of π into the identity permutation by exchanges of adjacent elements. For example, $3142 \rightarrow 1342 \rightarrow 1324 \rightarrow 1234$ is a three-step swap sorting of permutation 1234 . Write a greedy algorithm for swap sorting that uses the minimum number of swaps to sort a permutation.